

Project Design Phase Solution Architecture

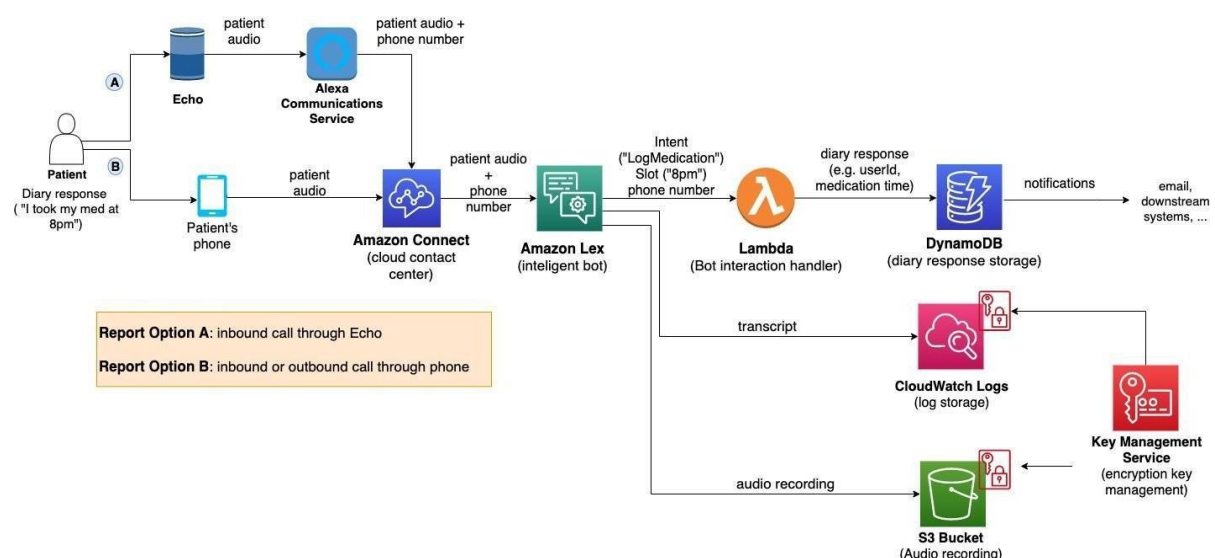
Date	15 May 2026
Team ID	HDI Predictor Team
Project Name	HDI Predictor using Machine Learning and Flask
Maximum Marks	4 Marks

Solution Architecture:

The HDI Predictor architecture connects a data-preparation notebook, a trained regression model, Flask route handlers, HTML templates, and JavaScript charts.

- The architecture moves data from the raw CSV file to the cleaned dataset, then to model training, saved inference, and finally the browser result page.
- The modular structure keeps model logic separate from presentation logic.
- The solution supports single predictions, country auto-fill, and CSV batch processing through the same core model.
- The architecture is designed for maintainability, testing, and a clear demonstration flow.

Reference: HDI Predictor project context from the SmartBridge handouts.



Template:

This document reflects the project workflow from training notebook to Flask application.

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